

Worksheet II

Introduction to Functions

Name: _____

Instructions

Use **pen**. Write questions and surprises in the margin in a different color. Some things in the reading are intentionally imprecise — find them.

Part 1 — What Is a Function?

[45 pts]

Reading

Read this section before attempting any problems.

The Definition

A **function** f from set A to set B , written $f: A \rightarrow B$, is a rule that assigns to each element of A exactly one element of B .

Think of a vending machine: you enter a code (input) and always receive exactly one item (output). A machine that sometimes gives you nothing, or sometimes gives two items for the same code, is broken. A function is a vending machine that is never broken.

Three things to name:

Term	Meaning
Domain (A)	The set of allowed inputs.
Codomain (B)	The set of declared outputs.
Range	The set of outputs actually produced.

The range is always a subset of the codomain, but they are not always equal. Most textbooks blur this distinction. We will not.

Notation

$f(x)$ is read “ f of x ” — the output of f when x is the input. It is **not** f times x .

Example. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = x^2$.

$$\begin{aligned}f(3) &= 9 \\f(-2) &= 4 \\f(0) &= 0\end{aligned}$$

Domain: $\mathbb{R}$. Codomain: $\mathbb{R}$. Range: all positive real numbers.

Injective Functions

A function f is **injective** (also called one-to-one) if different inputs always produce different outputs. Formally:

$$f(a) = f(b) \implies a = b$$

Is $f(x) = x^2$ injective? No. $f(2) = 4$ and $f(-2) = 4$, but $2 \neq -2$. Two different inputs produced the same output.

Is $f(x) = 2x + 1$ injective? Suppose $f(a) = f(b)$. Then $2a + 1 = 2b + 1$, so $2a = 2b$, therefore $a = b$. **Yes.**

Notice the proof structure: *suppose* the outputs are equal, derive that the inputs must be equal. This is your template.

Surjective Functions

A function $f: A \rightarrow B$ is **surjective** (also called onto) if every element of B is hit by at least one input. Formally:

$$\text{for every } b \in B, \text{ there exists } a \in A \text{ such that } f(a) = b$$

Is $f(x) = x^2$ surjective as $f: \mathbb{R} \rightarrow \mathbb{R}$? No: -1 is in the codomain but $f(x) = -1$ has no real solution.

What if we redefine the codomain as $[0, \infty)$? Then range equals codomain, so yes — surjective. Surjectivity depends on the codomain you declare.

Bijjective Functions

A function is **bijjective** if it is both injective and surjective. Every element of the codomain is hit by exactly one input — a perfect pairing. Only bijjective functions have inverses.

Foundation Practice

1. [15 pts] Let $f(x) = 3x - 2$. Fill in the table, then answer the questions below.

x	$f(x)$
0	
1	
-1	
4	
$\frac{1}{3}$	

- (a) What is $f(f(1))$? Answer: _____
- (b) Find x such that $f(x) = 0$. Answer: _____
- (c) Is f injective? Write a one-sentence proof using *suppose ... then ... therefore*.

2. [15 pts] Decide whether each rule defines a valid function from $A = \{1, 2, 3\}$ to $B = \{1, 2, 3, 4, 5\}$. If it does, state whether it is injective and/or surjective. If it does not, explain why not.

- (a) Each number maps to its square: $1 \mapsto 1$, $2 \mapsto 4$, $3 \mapsto 9$.

(b) Every number maps to 1: $1 \mapsto 1$, $2 \mapsto 1$, $3 \mapsto 1$.

(c) $1 \mapsto 2$, $2 \mapsto 4$, $2 \mapsto 5$.

3. [15 pts] For each function, state the domain, codomain, and range.

(a) $g: \mathbb{R} \rightarrow \mathbb{R}$ defined by $g(x) = x^2 + 1$.

(b) $h: \{1, 2, 3\} \rightarrow \{1, 4, 9\}$ defined by $h(x) = x^2$.

Is h surjective? Compare your answer to (a) and explain why they differ despite using the same formula.

Part 2 — Application

[35 pts]

4. [20 pts] From Bedrooms to Functions.

In Homework 1 Part 4, you were carpeting two Minecraft bedrooms. Both rooms are 10 blocks long. Bedroom 2 is 8 blocks wide. Bedroom 1 has unknown width w . Each carpet block takes 2 wool.

- (a) Write a function $f(w)$ giving the **total** wool needed for both rooms.

- (b) Evaluate $f(6)$. What does this number mean in context?

- (c) Find w such that $f(w) = 300$. What does this answer mean in context?

- (d) Is f injective? Write a formal proof using *suppose ... then ... therefore*.

(e) State the natural domain of f in the Minecraft context. Is f surjective onto the positive integers? Explain.

5. [15 pts] Speaking Functions.

(a) Read $f(x) = 3x^2 + 1$ as a complete English sentence. Write what you would say.

(b) A classmate says: “ $f(x)$ means f times x .” Are they right? Correct them in one sentence.

(c) Write $f(g(x))$ as a complete English sentence. What do you think this expression means?

Part 3 — Challenge

[20 pts]

6. [8 pts] The Finite Trap.

Try to build a function $f: \{1, 2, 3\} \rightarrow \{1, 2, 3\}$ that is injective but *not* surjective. Spend two minutes trying. Then explain what happens and why.

7. [12 pts] The Collatz Function.

Define $C: \mathbb{N} \rightarrow \mathbb{N}$ by:

$$C(n) = \begin{cases} n/2 & \text{if } n \text{ is even} \\ 3n + 1 & \text{if } n \text{ is odd} \end{cases}$$

Starting from any positive integer, apply C repeatedly.

- (a) Compute the sequence starting at $n = 6$. Stop when you reach 1.

- (b) Compute the first 15 terms starting at $n = 27$.

(c) Is C injective? Either prove it or find a counterexample.

(d) Write a conjecture about what always happens when you apply C repeatedly starting from any positive integer.

Part 4 — Notebook Artifact

(ungraded)

On the **left page** of your notebook, write:

1. Two **Question Compass** questions about the word *function*. Pick any two directions.
2. One **muddiest point** — the single thing you are least sure about after today.

ANSWER KEY — DO NOT DISTRIBUTE

Worksheet II — Introduction to Functions

Reading — Intentional Imprecision

The range of $f(x) = x^2$ is stated as “all positive real numbers.” This is wrong: $f(0) = 0$, and 0 is neither positive nor negative. The correct range is $[0, \infty)$ — all *non-negative* reals. See if either student catches it. If not, ask: “What is $f(0)$? Is that positive?”

Part 1 — Foundation

Q1. $f(x) = 3x - 2$

x	$f(x)$
0	-2
1	1
-1	-5
4	10
$\frac{1}{3}$	-1

(a) $f(1) = 1$, so $f(f(1)) = f(1) = 1$.

Note: $f(1) = 1$ is a fixed point of f — the input equals the output. Worth mentioning if a student notices it.

(b) $3x - 2 = 0 \implies x = \frac{2}{3}$

(c) Suppose $f(a) = f(b)$. Then $3a - 2 = 3b - 2$, so $3a = 3b$, therefore $a = b$. Injective. ✓

Q2.

(a) **NOT a valid function.** $3^2 = 9$, but $9 \notin B = \{1, 2, 3, 4, 5\}$. The output is outside the declared codomain.

This is the curveball. Many students will say “yes, it’s x^2 ” without checking whether 9 is in B . Socratic move: “What is 3^2 ? Is that in B ?”

(b) **Valid function.** NOT injective (1, 2, 3 all map to the same output). NOT surjective (2, 3, 4, 5 are never hit).

This is the constant function. Students may say “that can’t be a function because the output never changes.” Push back: does each input have exactly one output? Yes. That’s all a function requires.

(c) **NOT a valid function.** The element $2 \in A$ maps to both 4 and 5. One input, two outputs — broken vending machine.

Q3.

(a) $g(x) = x^2 + 1$, $g: \mathbb{R} \rightarrow \mathbb{R}$.

Domain: $\mathbb{R}$. Codomain: $\mathbb{R}$. Range: $[1, \infty)$.

(Since $x^2 \geq 0$, we have $x^2 + 1 \geq 1$. And every value ≥ 1 is achieved.)

(b) $h: \{1, 2, 3\} \rightarrow \{1, 4, 9\}$, $h(x) = x^2$.

Domain: $\{1, 2, 3\}$. Codomain: $\{1, 4, 9\}$. Range: $\{1, 4, 9\}$.

Yes, h is surjective — range = codomain.

Contrast with (a): same formula, but the codomain in (a) is all of $\mathbb{R}$, which $x^2 + 1$ never fully covers. In (b) the codomain was declared to be exactly the range, so surjectivity is automatic.

Key insight: surjectivity is a relationship between the function and the declared codomain. You can make any function surjective by shrinking the codomain to equal the range.

Part 2 — Application

Q4. Minecraft bedrooms.

(a) Wool for Bedroom 1: $2 \times 10 \times w = 20w$. Wool for Bedroom 2: $2 \times 10 \times 8 = 160$.

$$f(w) = 20w + 160$$

(b) $f(6) = 20(6) + 160 = 120 + 160 = \mathbf{280}$.

Meaning: if Bedroom 1 is 6 blocks wide, you need 280 wool total.

(c) $20w + 160 = 300 \implies 20w = 140 \implies w = 7$.

Meaning: if you want to collect exactly 300 wool, Bedroom 1 should be 7 blocks wide.

(d) Suppose $f(a) = f(b)$. Then $20a + 160 = 20b + 160$, so $20a = 20b$, therefore $a = b$. Injective. ✓

(e) Natural domain: positive integers ($w \geq 1$, whole blocks only).

Not surjective onto $\mathbb{Z}^+$. The outputs of f are 180, 200, 220, ... — only multiples of 20 starting at 180. Most positive integers (e.g. 181) are never hit.

Good Socratic question: “Is every even number a possible output? What about 182?” $20w + 160 = 182 \implies w = 1.1$ — not a whole number.

Q5. Speaking functions.

(a) “ f of x equals three x squared plus one.”

(b) No. $f(x)$ means “the output of function f when the input is x ” — it is not multiplication. The parentheses are function-application notation, not grouping.

(c) “ f of g of x .” Meaning: first apply g to x , then apply f to the result. This is function **composition** — a preview of what’s ahead.

Part 3 — Challenge

Q6. The Finite Trap.

It is impossible. Any injective function $f: \{1, 2, 3\} \rightarrow \{1, 2, 3\}$ must send three distinct inputs to three distinct outputs — but there are only three elements in the codomain, so all three must be used. Therefore it is automatically surjective.

On finite sets of equal size: **injective** $\iff$ **surjective** $\iff$ **bijective**.

The principle at work is the Pigeonhole Principle: if you have n pigeons and n holes, and each pigeon goes in a different hole, every hole has a pigeon. Name this for the students when they get stuck. This is one of the most useful ideas in all of combinatorics.

Q7. The Collatz Function.

(a) Starting at $n = 6$:

$$6 \rightarrow 3 \rightarrow 10 \rightarrow 5 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$$

8 steps.

(b) First 15 terms starting at $n = 27$:

$$27, 82, 41, 124, 62, 31, 94, 47, 142, 71, 214, 107, 322, 161, 484, \dots$$

The sequence from 27 takes 111 steps and reaches a maximum of 9232 before collapsing to 1. This is a great illustration of why checking examples is not the same as proving — it gets much worse before it gets better.

(c) C is **not injective**. Counterexample: $C(3) = 10$ and $C(20) = 10$, but $3 \neq 20$.

(Check: 3 is odd, $3(3) + 1 = 10$. 20 is even, $20/2 = 10$.)

Students who look for a counterexample by trying small cases will find this. Students who try to prove injectivity will get stuck — which is the right outcome. Ask: “Did you find a proof, or did you give up? Did you try to find a counterexample?”

(d) **Conjecture:** Starting from any positive integer, repeated application of C eventually reaches 1.

This is the Collatz Conjecture, open since 1937. Verified computationally for all n up to roughly 2^{68} . Paul Erdős said: “Mathematics is not yet ready for such problems.” If either student conjectures this correctly, tell them what it is.